Introduction to Applied Regression Analysis

ICPSR Summer Program, May 26–29, 2026

1 Course Information

Instructor	Patrick Shea
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Days and Time	May 26–29, 2026, 10:00 a.m.–3:30 p.m. (Eastern)

2 Course Description

This course is designed for individuals seeking to enhance their quantitative research capabilities, with or without previous experience in statistics. Our interactive sessions, bolstered by the use of R, promise a practical learning environment conducive to understanding and applying statistical analysis in real-world contexts. Tailored for social science researchers, this course offers a comprehensive exploration of statistical analysis techniques essential for interpreting research findings.

By the end of this course, participants will be equipped with the skills to design robust research studies, manage and analyze data proficiently, and draw meaningful conclusions from their findings. Whether you aim to contribute to academic discourse or apply these skills in the field, this course offers the foundation you need to excel.

The course will be centered around several main topics covering the basic analysis of ordinary least squares (OLS), estimating bivariate and multivariate regression models, assessing overall fitness of a regression equation and diagnostic tests, as well as exploring extensions beyond OLS including panel data models, instrumental variables, and logit/probit models for discrete outcomes.

The course adopts a 'learning by doing' approach. Through hands-on exercises and real-world examples using the statistical software R, participants will not only work on regression analysis but will also apply these concepts directly to social science research.

3 Pre-requisites

While it would be beneficial for participants to have a working knowledge of elementary statistical concepts, I do not assume that students have any special math background or extensive experience with statistics or statistical software. These concepts will be reviewed in the first session.

4 Statistical Software

This course uses **R**, a free, open-source statistical computing environment available for all major operating systems. The course will emphasize hands-on use of R for regression analysis, with guidance provided throughout. Example code will be provided and the instructor will show students how to responsibly use AI-assisted tools as a supplement for coding and debugging in R. This includes discussion of how to generate code using various AI-assisted techniques, verify AI-generated output, and understand the limitations of these tools for statistical analysis.

Students should download R and RStudio as soon as possible and ensure both programs are installed and ready to run before the first session. No prior experience with R is assumed.

To get started, students can consult introductory materials such as Rafael Irizarry's *Getting Started with R*.

Upon request, the instructor can also provide examples and supplemental resources using the statistical software **Stata**.

5 Readings and Textbooks

The main textbook for the course:

- Wooldridge, Jeffrey M. *Introductory Econometrics: A Modern Approach.* 5th edition (earlier/later editions are fine).

The following is a *recommended* text for the course:

- Gelman, Andrew, Jennifer Hill, and Aki Vehtari. *Regression and Other Stories.* Cambridge University Press, 2020.

6 Preliminary Schedule

Each of the four days will be broken up into two sessions, with an extended lunch break between each session. Each session will comprise a lecture component and an R lab exercise. Time will be provided for questions and for working on examples in R.

6.1 Day 1

6.1.1 Morning Session: Foundations of Statistics and Regression

- Welcome and Introductions
- Statistics Review (Wooldridge Appendix B & C and Ch. 1)
  - Models: translating research questions into formal structures
  - Data and the role of variation
  - Estimands: defining what we want to estimate
  - Center and spread of distributions: mean, median, variance, and standard deviation
  - Residuals as deviations from predictions
  - The role of randomness and uncertainty in inference
- R Lab #1

6.1.2 Afternoon Session: Introduction to OLS

- Ordinary Least Squares (Wooldridge Ch. 2)
  - Covariance and correlation
  - The OLS slope: from correlation and variance to regression coefficients
  - The Conditional Expectation Function (CEF) decomposition
  - Residuals and the assumption $E[\varepsilon|X] = 0$
  - R^2: decomposing total variance into explained and unexplained
- R Lab #2

6.2 Day 2

6.2.1 Morning Session: Using OLS

- Interpreting and Testing Regression Results (Wooldridge Chs. 3 and 4)
  - Interpreting coefficients: sign, substance, and statistical significance
  - Reading regression output: coefficients, standard errors, t-values, p-values, R^2
  - Null hypothesis testing and the Type I/II error tradeoff
  - Sampling distributions and standard errors
  - Confidence intervals
  - Presenting results with tables and graphical displays
- R Lab #3

6.2.2 Afternoon Session: Interpretation and Multivariate Regression

- Multivariate regression and Simpson's Paradox (Wooldridge Chs. 4 and 7)
  - Extending regression to multiple predictors
  - Dummy variables in OLS
  - Coefficient interpretation with multiple regressors
- R Lab #4

6.3 Day 3

6.3.1 Morning Session: Model Specification

- Specification (Wooldridge Ch. 6)
  - Variable selection and model design
  - Functional form choices: linear-log, log-linear, log-log, and polynomial specifications
  - Interaction effects
  - Lagged variables and dynamic specifications
- R Lab #5

6.3.2 Afternoon Session: Potential Problems in Multiple Regression Analysis

- The Classical Model (Wooldridge Chs. 3 and 10.3)
  - The Gauss-Markov assumptions and the Best Linear Unbiased Estimator (BLUE)
  - Multicollinearity and variance inflation factors (VIF)
  - Outliers, leverage, and influence
  - Heteroskedasticity (Wooldridge Ch. 8)
  - Serial correlation (Wooldridge Ch. 12)
  - Diagnostic tests and residual analysis
- R Lab #6

6.4 Day 4

6.4.1 Morning Session: Topics in Regression Analysis

- Panel-Data Models (Wooldridge Chs. 13 and 14)
  - Pooled OLS regressions
  - Fixed-effect models
  - Random-effect models
- Two-Stage Models
  - Directed Acyclic Graphs (DAGs) for causal reasoning
  - Mediation analysis
  - 2SLS and instrumental variables (Wooldridge Ch. 15)
- R Lab #7

6.4.2 Afternoon Session: Generalized Linear Models

- Models of Discrete and Limited Dependent Variables (Wooldridge Ch. 17)
  - The Linear Probability Model (LPM)
  - The Logit and Probit models
  - When to use LPM vs. Logit/Probit
- R Lab #8